Unit-5

Random Variable and Mathematical Expectation

Assignment no. 1

Discrete Random Variables

1. Priyanka works in an automotive tire factory. The number X of sound but blemished tires that he produces on a random day has the probability distribution

X	2	3	4	5
P (X=x)	0.48	0.36	0.12	0.04

a. Find the probability that Priyanka will produce more than three blemished tires tomorrow.

b. Find the probability that Priyanka will produce at most two blemished tires tomorrow.

c. Compute the CDF of F (x).

2. A pair of fair dice is rolled. Let X denote the sum of the number of dots on the top faces. (a) Construct the probability distribution of X. (b) Find $P(X \geq 9)$. (c) Find the probability that X takes an even value.

[Ans: (b) 0.27, (c) = 0.5]

3. A fair coin is tossed three times, and the random variable is defined as the number of heads found. Describe a sample space for the experiment. Write down the probability function of , and sketch its probability function and cumulative distribution function.

Jay Prakash Kapari

4. A discrete random variable X has the following probability distribution:

X	- 1	0	1	4
P (x)	0.2	0.5	a	0.1

(i) Find the value of a.

(ii) Find P (0).

(iii) Find P(X > 0).

(iv) Find P (X ≥ 0).

(v) Find P (X≤−2).

(vi) Find F (1) and F (x).

(vii) Compute the mean μ of X.

(viii) Compute the variance σ^2 of X.

(ix) Compute the standard deviation σ of X.

[Ans: 0.2, 0.5, 0.3, 0.8, 0, 0.9, 0.4, 1.84, 1.3565]

5. Tybalt receives in the mail an offer to enter a national sweepstakes. The prizes and chances of winning are listed in the offer as: $5 million, one chance in 65 million; $150,000, one chance in 6.5 million; $5,000, one chance in 650,000; and $1,000, one chance in 65,000. If it costs Tybalt 44 cents to mail his entry, what is the expected value of the sweepstakes to him?

6. A random variable X has following probability function.

Value of X =x	- 2	- 1	0	1	2	3
P (X =x)	k	k	2k	2k	3k	k

a. Find the value of k.
b. Find P (X<2), P (- 2<X<2), P (0≤X<2), P (X≥-2)
c. Find F (-2), F (0), F (3) and
d. Construct the CDF. Of F (x)

7. In a given business venture a man can make a profit of Rs 1000 with probability 0.8 or suffers a loss of Rs. 400 with probability 0.2. Determine his expectation.

8. A random variable X has following probability mass function

X	0	1	2	3	4	5	6
P (x)	k	3k	5k	7k	9k	11k	13k

Find (i) k, (ii) P (X < 3), P (X ≥ 3), (iii) E (X), E (X^2), E (X^3), E (X^4), and (iv) mean and variance of X.

9. A discrete random variable X has probability mass function

Defects (X = x)	0	1	2	3	4	5
Number of pages	0.0933	0.1333	0.2667	a	0.16	0.1067

Jay Prakash Kapari

(i) Find the value of a. (ii) Find first four moments about origin. (iii) Find first four central moments about mean. (iv) Measure of central tendency, dispersion, skewness and kurtosis.

Continuous Random Variables

1. The installation time, in hours, for a certain software module has a probability density function f(x) = k ($1-x^3$) for 0 < x < 1. Find k and compute the probability that it takes less than 1/2 hour to install this module.

2. Suppose the distance time X between a point target and a shot aimed at the point in a coin operated target game is continuous random variable with probability density function:

$$f(x) = \begin{cases} \frac{3}{4}(1-x^2), & -1 < x < 1 \\ 0, & otherwise \end{cases}$$

i) Compute P (x>0)

ii) Compute P (-0.5<x<0.5)

iii) Compute P (x<-0.25 or x>0.25)

3. A random variable X has density function

$$f(x) = \begin{cases} ce^{-3x}; & x > 0 \\ 0\ ; & x \leq 0 \end{cases}$$

Find (a) the constant c, (b) P (1 < x < 2), (c) P (x ≥ 3), (d) P (X < 1), F (x).

4. The distribution function of a random variable X is given by

$$F(x) = \begin{cases} cx^2; & 0 \le x < 3 \\ 1 \ ; & x \ge 3 \\ 0 \ ; & x < 0 \end{cases}$$

Find (a) the constant c, (b) the density function, (c) P(X < 1), (d) P(1 < X < 2).

5. A continuous random variable has probability density function

$$f(x) = \begin{cases} ax & ; \ if \ 0 \le x < 1 \\ a & ; \ if \ 1 \le x < 2 \\ 3a - ax & ; \ if \ 2 \le x < 3 \\ 0 & ; \ elsewhere \end{cases}$$

(i) Find the value of a. (ii) Find P (1.5 < x < 2.5). (iii) Find F (1.5).

6. A continuous random variable x has a pdf

$$f(x) = \begin{cases} kx^4; & 0 < x < 1 \\ 0 \ ; & Otherwise \end{cases}$$

(a) Find the value of k. (b) Find the value of m if P (x ≤ m) = P (x > m). (c) Find the value of n if P (x > n) = 0.5.

7. The distribution function of x is

Jay Prakash Kapari

$$f(x) = \begin{cases} a + bx^2; & 0 \le x \le 1 \\ 0 & ; \ elsewhere \end{cases}$$

If E (X) = $\frac{3}{5}$, find the value of a and b.

8. Find the mean and variance of the continuous distribution with probability density

$$f(x) = \begin{cases} 12x^2(1-x); & 0 \le x \le 1 \\ 0 & ; \ elsewhere \end{cases}$$

9. Find the mean and variance of the continuous distribution with probability density

$$f(x) = \begin{cases} kx^2(1-x); & 0 \le x \le 1 \\ 0 & ; \ elsewhere \end{cases}$$

(i) Find the value of k. (ii) Find expected value of X. (iii) Find $E(X^2), E(X^3), E(X^4)$. (iv) Find mean and variance.

10. Find the mean and variance of the continuous distribution with probability density

$$f(x) = \begin{cases} kx^3 e - x & ; \ x \ge 0 \\ 0 & ; \ elsewher \end{cases}$$

(i) Find the value of k. (ii) Find first four moments about origin. (iii) Find first four central moments about mean. (iv) Measure of central tendency, dispersion, skewness And kurtosis.

11. Find first four moments about origin, first four moments about mean, measure of central tendency measure of dispersion, measure of skewness and measure of kurtosis and interpretation from following distribution.

$$f(x) = \begin{cases} kx(1-x)^2; & 0 \le x \le 1 \\ 0 & ; \quad elsewhere \end{cases}.$$

Jay Prakash Kapari